FootfallCam AI Evaluation — Staff Identification

Clip: `sample.mp4`, overhead fish-eye 3D sensor, 960×720, 25 fps, 1341 frames (53.6 s). **Tasks:** (1) which frames contain the staff member (lanyard name-tag wearer); (2, bonus) their `(x, y)` in each.

Answer

Task 1 — every frame (0–1340, 0.0–53.6 s). The staff member is on camera for the whole clip; they only move between two desks, they never leave. So the work is *locating* them, not deciding when they are present.

Task 2 — trajectory (full per-frame data in `outputs/staff_frames.csv`):

time	frames	location	approx (x, y)
0:00–0:13	0–320	seated at the **left-wall desk**, light-blue shirt	105, 415
0:13–0:18	320–460	stands, walks **up the central corridor**	→ 525, 170
0:18–0:22	460–555	around the **top / top-right desks**	505, 120
0:22–0:26	555–650	**back down** to the **bottom-centre desk**; bends over it; **puts a hoodie on**	445, 600
0:26–0:52	650–1300	around the bottom-centre desk, then walks back to the left	→ 100, 410
0:52–0:53.6	1300–1340	back at the **left-wall desk**, in the hoodie	95, 400

The staff wears a **light-blue long-sleeve shirt** for the first ~26 s, then a **dark hoodie** — which is why the brief's two overhead snapshots (one light top, one dark hoodie with the card hanging) are the same person at different times.

Method

1. **Motion.** MOG2 background subtraction each frame → cleaned foreground blobs, constrained to the auto-detected fish-eye disc. Seated people barely register; a walker leaves a large travelling blob.
2. **Propose the mover.** Blobs are linked frame-to-frame by nearest centroid; the tracklet that travels furthest is the candidate staff member. This alone recovers the corridor walk (0:13–0:26).
3. **Waypoints → full track.** Eight positions of the staff, verified against the clip (`solution/sample_keyframes.txt`, `frame:x,y,w,h`), anchor the track. The tool interpolates between them and snaps each frame to the nearest colour-consistent motion blob, so the box follows real movement.
4. **Name-tag cue.** Inside the tracked box, look for the lanyard card — a small, bright, near-white local brightness peak on the upper body. Logged per frame as `tag_score`; corroborating evidence, not a gate (the card is only intermittently visible from straight above).

Fish-eye → floor (bonus): equidistant model `r_pixel ∝ θ`, ground radius `= mount_height·tanθ`, bearing preserved. Mount height unknown, so `x_floor, y_floor` are in "ceiling heights" — enough to show the path shape (`trajectory_floor.png`).

Why it is analyst-assisted, not fully automatic

Fully-automatic single-target tracking does not survive this clip: ~20 bodies in a distorted overhead view, the subject stationary at a desk most of the time, a mid-clip change of clothes, and a *second* dark-clothed colleague sharing the bottom-centre desk after 0:26. A YOLOv8 + ByteTrack pass (`--backend yolo`, included) fragments into ~50 short tracklets with constant ID switches. Eight verified clicks beat an automatic tracker that then guesses; the pipeline supports both — auto proposal when one person clearly moves, `--keyframes` for hard clips.

Confidence

- **Solid** — present the whole clip; start and end at the left-wall desk; the corridor walk and its timing; the clothing change.
- **Soft** — exact position 0:26–0:52 (two dark-clothed people at one desk); that leg is interpolated between two waypoints, so it draws as a straight diagonal rather than the true route.
- **Next** — a person re-ID embedding (e.g. OSNet) instead of the HSV histogram would separate the two dark figures and remove most manual waypoints; sensor intrinsics + mount height would make the `*_floor` columns metric.

Run it

```
cd solution
python -m pip install -r requirements.txt          # numpy, opencv-python, matplotlib

# provided clip (uses the verified waypoints)
python detect_staff.py --video "<path to sample.mp4>" --out ../outputs \
       --keyframes-file sample_keyframes.txt

# a fresh test clip, fully automatic
python detect_staff.py --video other.mp4 --out ../outputs

# a fresh test clip, with a few waypoints of the staff
python detect_staff.py --video other.mp4 --out ../outputs \
       --keyframes "5:70,370,70,90; 400:520,120,80,150; 1200:90,400,70,95"
```

Outputs: `staff_frames.csv`, `annotated.mp4`, `timeline.png`, `trajectory.png`, `trajectory_floor.png`, `tag_evidence.png`, `summary.json`. All tunables are in one `P = dict(...)` at the top of `detect_staff.py`.